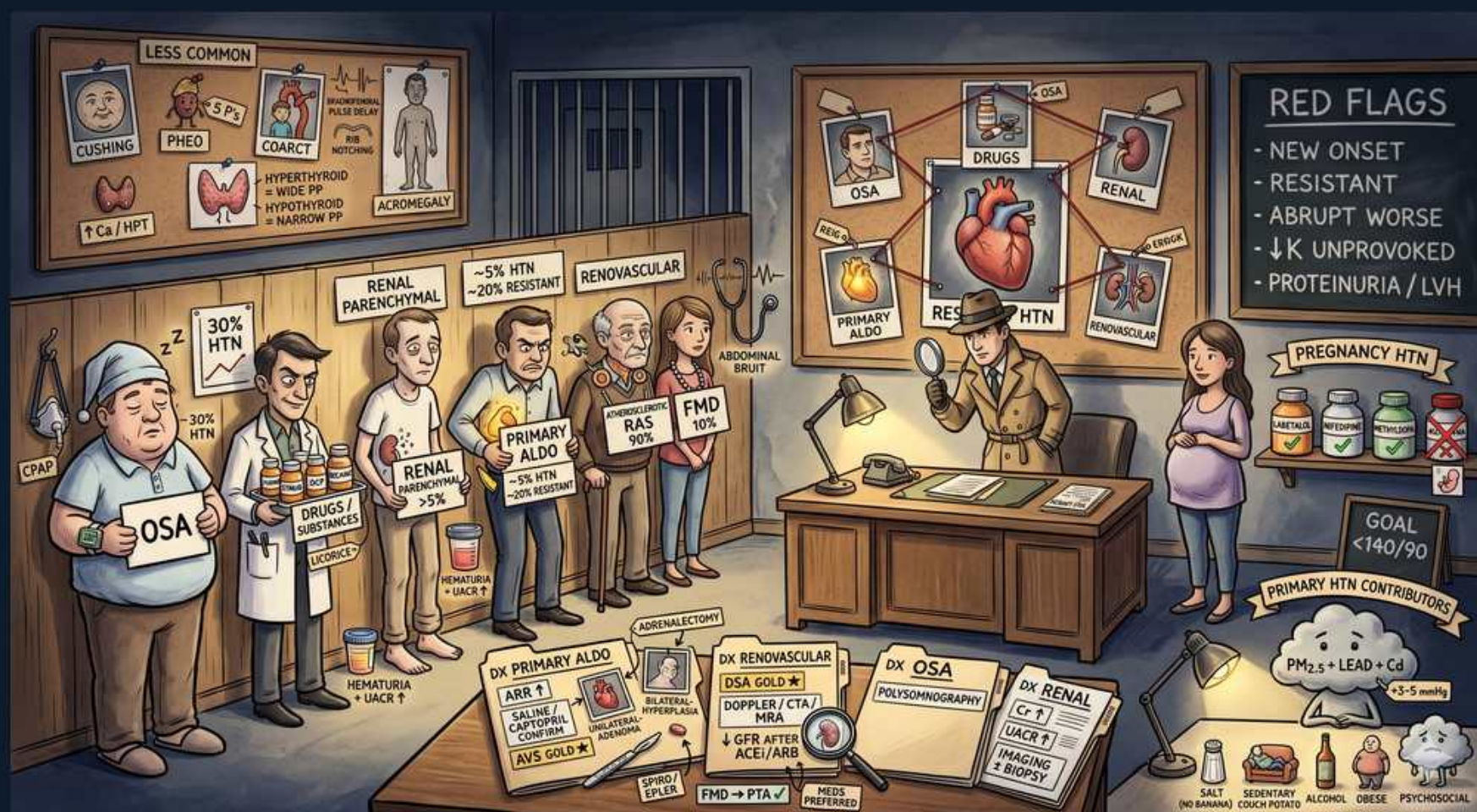


Hypertension — Secondary HTN (Causes, Red Flags & Workup)



1. Red flags to screen

WHAT YOU SEE

RED FLAGS chalkboard: new onset, resistant, abrupt worse, low K, proteinuria/LVH

WHAT IT ENCODES

Screen for secondary HTN when red flags are present: onset before age 30 or after 55, resistant HTN (uncontrolled on ≥ 3 drugs including a diuretic), abrupt worsening of previously controlled HTN, unprovoked/diuretic-induced hypokalemia, proteinuria or LVH out of proportion, or a hypertensive emergency at presentation.

BOARD TRAP

WRONG answer = order a full secondary workup on a typical 50-year-old with mild essential HTN and no red flags. >90% of adult HTN is primary; indiscriminate screening yields false positives. The discriminator is the PRESENCE of red flags, not the diagnosis of HTN itself.

2. Primary aldosteronism

WHAT YOU SEE

A suspect in the police lineup holding the PRIMARY ALDO placard beside an adrenal gland, tagged ~5-20% of resistant HTN — lined up as the top endocrine/identifiable secondary culprit.

WHAT IT ENCODES

Primary aldosteronism (Conn syndrome) is the most common ENDOCRINE and most common identifiable secondary cause, found in ~5–20% of resistant HTN. Caused by bilateral adrenal hyperplasia (most common) or an aldosterone-producing adenoma. Suspect with HTN + spontaneous or easily provoked hypokalemia, or resistant HTN even with NORMAL potassium.

BOARD TRAP

WRONG answer = exclude primary aldosteronism because potassium is normal. Most patients with primary aldosteronism are now NORMOkalemic; hypokalemia is neither sensitive nor required. Resistant HTN alone warrants screening with the aldosterone-to-renin ratio.

3. Aldo screening test

WHAT YOU SEE

The DX PRIMARY ALDO papers on the workup desk: ARR screen → confirm → AVS marked GOLD — the step ladder pictures screen, then confirm, then lateralize before surgery.

WHAT IT ENCODES

Diagnostic sequence for primary aldosteronism: (1) SCREEN with plasma aldosterone-to-renin ratio (ARR) — high aldosterone + suppressed renin; (2) CONFIRM with a salt-loading/saline suppression or fludrocortisone suppression test (aldosterone fails to suppress); (3) LATERALIZE with adrenal vein sampling (AVS, the gold standard) to distinguish unilateral adenoma (→ adrenalectomy) from bilateral hyperplasia (→ MRA).

BOARD TRAP

WRONG answer = go straight to adrenalectomy based on a CT showing an 'adenoma.' CT cannot reliably distinguish a functional adenoma from a non-functioning incidentaloma or from bilateral hyperplasia — AVS is required to lateralize before surgery. Also: stop MRAs/eplerenone ~4–6 weeks before measuring the ARR (they invalidate it).

4. Low K + low renin

WHAT YOU SEE

The primary-aldo suspect's clue card showing low K+ with suppressed renin — the paired low-K, low-renin labs are the Conn fingerprint.

WHAT IT ENCODES

Primary aldosteronism produces HIGH aldosterone with LOW (suppressed) renin, plus hypokalemia and metabolic alkalosis (aldosterone drives renal K+ and H+ loss via ENaC/principal cells). The suppressed renin is the key lab signature distinguishing it from secondary (renin-driven) causes.

BOARD TRAP

WRONG answer = diagnose primary aldosteronism when BOTH renin and aldosterone are HIGH. High-renin + high-aldosterone points to a SECONDARY hyperaldosteronism such as renovascular HTN or a renin-secreting tumor, not Conn syndrome. The discriminator is the renin level: low = primary, high = secondary.

5. Renal artery stenosis

WHAT YOU SEE

A lineup suspect with the RENOVASCULAR / RAS 90% placard and a stethoscope on the belly for an abdominal bruit — the listening doctor cues the renovascular bruit.

WHAT IT ENCODES

Renal artery stenosis causes renovascular HTN: reduced renal perfusion → renin release → angiotensin II and aldosterone-driven HTN. Classic clues: an abdominal/flank systolic-diastolic bruit, recurrent flash pulmonary edema, and a notable rise in creatinine after starting an ACEi/ARB.

BOARD TRAP

WRONG answer = attribute every abdominal bruit to RAS, or miss RAS because the bruit is absent (bruit is specific but insensitive). Key board clue: an acute creatinine rise after RAAS blockade, or asymmetric kidney size on imaging.

6. FMD vs atherosclerosis

WHAT YOU SEE

FMD young woman vs atherosclerotic older

WHAT IT ENCODES

Two causes of RAS by demographic: FIBROMUSCULAR DYSPLASIA — young women (<50), affects the mid-to-distal renal artery with a 'string-of-beads' angiographic appearance, treated with angioplasty; ATHEROSCLEROTIC RAS — older patients with vascular risk factors, affects the proximal/ostial artery, managed primarily with medical therapy.

BOARD TRAP

WRONG answer = stent an atherosclerotic RAS lesion routinely to control BP. The CORAL/ASTRAL trials showed stenting does NOT improve outcomes over optimal medical therapy for atherosclerotic RAS. Angioplasty IS appropriate for FMD. Discriminator: age/sex and lesion location/morphology.

7. ACEi after RAS warning

WHAT YOU SEE

The DX RENOVASCULAR papers: Doppler / CTA / MRA imaging plus a 'GFR after ACEi/ARB' warning — flags the creatinine jump after RAAS blockade in bilateral RAS.

WHAT IT ENCODES

Workup for renovascular HTN: noninvasive imaging with Doppler ultrasound, CTA, or MRA (catheter angiography is the gold standard but invasive). In RAS the efferent arteriole depends on angiotensin II to maintain GFR; starting an ACEi/ARB removes that tone and can precipitate AKI — especially in BILATERAL RAS or stenosis of a solitary kidney.

BOARD TRAP

WRONG answer = continue the ACEi/ARB and ignore a sharp creatinine rise after starting it. A >30% creatinine rise (or AKI) after RAAS blockade is a red flag for bilateral RAS — stop the drug and evaluate. Conversely, a small (<30%) rise in unilateral disease is expected and acceptable.

8. Pheochromocytoma

WHAT YOU SEE

The PHEO suspect with an adrenal-medulla icon and an episodic-spikes BP graph — the spiky tracing pictures paroxysmal catecholamine surges.

WHAT IT ENCODES

Pheochromocytoma: catecholamine-secreting tumor of the adrenal medulla → EPISODIC (paroxysmal) HTN with the classic triad of headache, palpitations, and diaphoresis. Screen with plasma free metanephrines or 24-hour urine metanephrines/catecholamines; localize with CT/MRI. Associated with MEN2, VHL, and NF1. Rule of 10s.

BOARD TRAP

WRONG answer = start a beta-blocker first to control the tachycardia/HTN. Unopposed alpha stimulation precipitates a hypertensive crisis. ALWAYS give alpha-blockade (phenoxybenzamine) first, then add a beta-blocker — same pre-operative sequence. Also WRONG: screening with plasma catecholamines (less sensitive) instead of metanephrines.

9. Cushing

WHAT YOU SEE

A round, moon-faced suspect holding the CUSHING placard with central-obesity/striae features — the cushingoid body habitus is drawn on the figure.

WHAT IT ENCODES

Cushing syndrome: cortisol excess → HTN (cortisol has mineralocorticoid activity and potentiates catecholamines) with central obesity, moon facies, purple abdominal striae, proximal muscle weakness, easy bruising, and hyperglycemia. Screen with late-night salivary cortisol, 24-hr urinary free cortisol, or low-dose dexamethasone suppression test.

BOARD TRAP

WRONG answer = order a single random serum cortisol to diagnose Cushing. Random cortisol is unreliable due to diurnal variation and stress; use one of the three validated screening tests, then ACTH and high-dose dexamethasone to localize (pituitary vs ectopic vs adrenal).

10. Coarctation

WHAT YOU SEE

The COARC suspect showing a wide arm pulse but weak/delayed leg pulses and CXR rib notching — the arm-vs-leg pulse mismatch is the coarctation tell.

WHAT IT ENCODES

Coarctation of the aorta: narrowing distal to the left subclavian → upper-extremity HTN with weak/delayed femoral pulses (radio-femoral delay) and an arm-leg systolic BP gradient (arm > leg). CXR shows rib notching (collateral intercostal arteries) and the '3 sign'. Associated with bicuspid aortic valve and Turner syndrome.

BOARD TRAP

WRONG answer = work up a young hypertensive patient for endocrine causes while ignoring the legs. Always check femoral pulses and four-limb BPs in young HTN; diminished/delayed femoral pulses point to coarctation. Confirm with echo/CT/MR angiography, not just CXR.

11. OSA

WHAT YOU SEE

A sleeping man with 'Zzz' holding the OSA / ~30% HTN placard — the snoozing figure pictures obstructive sleep apnea driving resistant HTN.

WHAT IT ENCODES

Obstructive sleep apnea is a very common and underrecognized cause of secondary and resistant HTN (sympathetic activation, nocturnal hypoxia, non-dipping BP pattern). Screen snorers, obese patients, and resistant hypertensives; confirm with polysomnography. CPAP improves BP modestly, especially in resistant HTN.

BOARD TRAP

WRONG answer = overlook OSA in a resistant hypertensive because they 'just snore.' OSA is one of the most frequent contributors to resistant HTN; daytime somnolence (Epworth), witnessed apneas, and a non-dipping nocturnal BP profile are key clues.

12. Renal parenchymal

WHAT YOU SEE

A suspect with the RENAL PARENCHYMAL placard showing hematuria and low GFR — drawn as the most common secondary cause overall (CKD).

WHAT IT ENCODES

Renal parenchymal disease (chronic kidney disease — diabetic nephropathy, glomerulonephritis, polycystic kidney disease) is the MOST COMMON secondary cause of HTN overall (vs primary aldosteronism, the most common ENDOCRINE cause). Clues: elevated creatinine/low GFR, proteinuria, hematuria, and abnormal renal ultrasound.

BOARD TRAP

WRONG answer = confuse 'most common secondary cause overall' (renal parenchymal/CKD) with 'most common endocrine/identifiable secondary cause' (primary aldosteronism). Boards test this distinction directly. ACEi/ARB are first-line for proteinuric CKD-related HTN.

13. Drugs cause HTN

WHAT YOU SEE

A suspect clutching a fistful of pill bottles labeled DRUGS / SUBSTANCES — the bottles picture NSAIDs, OCPs, decongestants, licorice etc. as a reversible cause.

WHAT IT ENCODES

Drug- and substance-induced HTN: NSAIDs (Na⁺ retention, blunt antihypertensives), oral contraceptives/estrogen, decongestants (pseudoephedrine), glucocorticoids, stimulants (cocaine, amphetamines), VEGF inhibitors, calcineurin inhibitors, excess alcohol, and licorice (mimics mineralocorticoid excess — pseudohyperaldosteronism with low renin AND low aldosterone).

BOARD TRAP

WRONG answer = escalate antihypertensives without taking a thorough medication/substance history. A careful review (including OTC NSAIDs, decongestants, supplements, and licorice) often reveals a reversible cause. Licorice classically gives the labs of aldosteronism but with SUPPRESSED aldosterone.

14. Pregnancy HTN

WHAT YOU SEE

Pregnant woman, PREGNANCY HTN, goal <140/90

WHAT IT ENCODES

Pregnancy-related HTN: chronic HTN, gestational HTN (after 20 weeks, no proteinuria), and preeclampsia (HTN + proteinuria/end-organ damage). Safe agents are labetalol, nifedipine (extended-release), and methyldopa; treat to roughly <140/90. Add magnesium sulfate for severe preeclampsia/eclampsia seizure prophylaxis.

BOARD TRAP

WRONG answer = continue or start an ACEi/ARB (or a direct renin inhibitor) in a pregnant patient. They are teratogenic (fetal renal dysgenesis, oligohydramnios, skull defects) and absolutely contraindicated, especially in the 2nd/3rd trimesters. Switch to labetalol/nifedipine/methyldopa before conception.